

ISRC Robotics Event: Maze Solver Rules, Regulations, and Judging Criteria

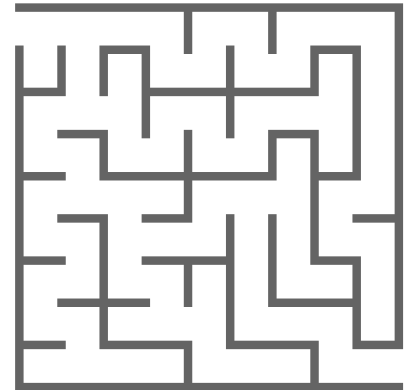
Rules and Regulations

1. Eligibility

- **Age Groups:**
 - **Group 1:** 15-18 years
 - **Group 2:** 19-22 years
- **Team Size:** Teams must consist of 2-5 members.

2. Event Structure

- **Rounds:**
 - **First Round:** All teams will participate.
 - **Second Round:** All teams will participate.
 - **Final Round:** The top-performing teams from the second round will advance to the final round, featuring the most challenging maze.
- **Attempts:** Each team will be given two attempts per round. The best attempt (shortest time with the fewest penalties) will be considered for scoring.



3. Maze Specifications

- **Maze Design:** The maze will consist of straight paths, sharp turns, dead ends, and multiple routes to the exit, with increasing complexity in each round.
- **Path Width:** The path width will be consistent across all age groups at 10-30 cm, challenging participants to design compact and efficient robots.
- **Maze Size:**
 - **Group 1 (15-18 years):** 3x3 meter maze with moderate complexity.
 - **Group 2 (19-22 years):** 4x4 meter maze with higher complexity, including more dead ends and tighter turns.

4. Robot Requirements

- **Autonomous or Remote-Controlled:** Robots may either operate autonomously or be controlled remotely.
- **Size Restrictions:** Robots must fit within a 20x20 cm footprint.
- **Sensor Usage:** Robots are encouraged to use sensors like IR, ultrasonic, or cameras for maze navigation.
- **Power Supply:** Robots must be powered by batteries. Teams should have spare batteries on hand.

5. Competition Format

- **Time Trials:** The robot's performance is timed, and the fastest time to complete the maze with the fewest penalties is recorded.
- **Obstacle Navigation:** Robots must navigate the maze without colliding with walls or getting stuck. Deviating from the path or failing to solve the maze will incur penalties.

- ❑ **Restarts:** Teams are allowed to restart their robot if it fails to navigate the maze, but each restart will incur a time penalty.

6. Scoring and Penalties

- ❑ **Time Scoring:** The robot with the shortest time and fewest penalties in each round will receive the highest score.
- ❑ **Penalty Points:**
 - **Colliding with a wall:** +5 seconds
 - **Getting stuck:** +10 seconds
 - **Restarting the robot:** +15 seconds
- ❑ **Final Scoring:** The best score from the two attempts in each round will be considered for advancing to the next round and for final scoring.

7. Safety Regulations

- ❑ **Robot Safety:** Robots must be designed safely with no sharp edges or hazardous materials.
- ❑ **Maze Safety:** The maze area must be kept clear of all participants and spectators during the runs.
- ❑ **Emergency Stop:** Each robot must have an accessible emergency stop mechanism.

Judging Criteria for Maze Solver Event

Scoring Out of 1000 Points:

1. Innovation and Originality (200 points)

- ❑ **Creative Navigation Techniques (80 points):** Evaluate the originality of the robot's design and approach to solving the maze, including the use of innovative sensors or algorithms.
- ❑ **Advanced Problem-Solving (60 points):** Assess the robot's ability to handle unexpected challenges or deviations in the maze.
- ❑ **Novelty in Design (60 points):** Consider the uniqueness of the robot's physical design and its potential to introduce new concepts in robotics.

2. Technical Mastery (300 points)

- ❑ **Precision in Maze Navigation (120 points):** Evaluate the accuracy with which the robot navigates the maze, maintaining consistent direction and speed.
- ❑ **Sensor Integration (100 points):** Assess the effectiveness of the sensors used for maze detection and navigation.
- ❑ **Robustness and Durability (80 points):** Evaluate the robot's ability to function reliably under various conditions within the maze.

3. Practical Applicability and Global Impact (150 points)

- ❑ **Real-World Applications (70 points):** Consider the potential of the robot's design to be applied in real-world scenarios, such as search and rescue operations or automated navigation systems.
- ❑ **Environmental Considerations (50 points):** Evaluate the sustainability of the robot, including energy efficiency and material usage.
- ❑ **Scalability (30 points):** Assess the potential for the robot's design to be scaled for larger or more complex applications.

4. Presentation Excellence (200 points)

- ❑ **Technical Explanation (100 points):** Evaluate the team's ability to explain the technical aspects of their robot, including sensor integration, programming, and design choices.
- ❑ **Demonstration Quality (70 points):** Assess the effectiveness of the live demonstration, including the robot's performance and the clarity of the team's presentation.
- ❑ **Team Communication (30 points):** Consider how well the team collaborates and communicates their ideas during the presentation.

5. Judges' Challenge (100 points)

- ❑ **Real-Time Problem Solving (50 points):** Present a real-time challenge to the teams, such as modifying the maze or introducing a new obstacle, and evaluate their ability to adapt.
- ❑ **Cross-Disciplinary Collaboration (30 points):** Assess how well the team integrates knowledge from different STEM fields, such as engineering, programming, and design.
- ❑ **Time-Constrained Innovation (20 points):** Evaluate the team's ability to innovate under time constraints during the challenge.

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6. Robot Design and Functionality (50 points)

- ❑ **Mechanical Design (20 points):** Assess the structural design, including the materials used, compactness, and overall build quality of the robot.
- ❑ **Programming and Control (20 points):** Evaluate the robot's programming logic, control algorithms, and efficiency in executing tasks.
- ❑ **Adaptability (10 points):** Consider the robot's ability to adapt to different maze conditions, including speed adjustments and obstacle avoidance.